

YUEPENG HU

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EDUCATION

Duke University

PhD in Electrical and Computer Engineering
4th Year, PhD Candidate

Aug. 2022 - Present

Overall GPA: 3.73/4.0

Duke University

Master of Science in Electrical and Computer Engineering

Aug. 2022 - Jul. 2025

University of Science and Technology of China (USTC)

Bachelor of Science in Data Science

Aug. 2018 - Jun. 2022

Overall GPA: 3.86/4.3

KEYWORDS

AI Security, Generative AI, LLM Agent, MLLM, Watermarking

WORK EXPERIENCE

Meta Superintelligence Labs

Research Scientist Intern

Bellevue, WA

May 2026 - Present

- Studied reward-hacking behavior in RL-trained coding agents, focusing on how models represent problematic shortcuts in their reasoning
- Designed controlled single-agent and multi-agent experiments to measure when these behaviors emerge and how they propagate across agents
- Built evaluation pipelines for coding tasks, reasoning-trace analysis, and automated detection of reward-hacking patterns

TikTok Privacy Innovation Labs

Research Scientist Intern

San Jose, CA

May 2025 - Aug. 2025

- Proposed and formalized a novel cross-server prompt injection attack targeting LLMs integrated with Model Context Protocol (MCP)
- Demonstrated how malicious tool descriptions can influence LLM behavior across trusted services
- Developed and experimentally validated attack cases causing unauthorized actions across trusted services, such as data leakage via indirect tool invocation

RESEARCH EXPERIENCE

MalTool: Malicious Tool Attacks on LLM Agents

Supervised by Prof. Neil Gong

Durham, US

Sep. 2025 – Feb. 2026

- Proposed the first systematic study of code-level malicious tool attacks in LLM agent ecosystems and developed a CIA-based taxonomy of tool behaviors
- Built MalTool, a coding-LLM-based framework with execution-aware verification to automatically synthesize functionally correct and structurally diverse malicious tools
- Constructed large-scale datasets (6K+ tools) and demonstrated that existing commercial and agent-specific detectors fail to reliably identify such attacks

Fingerprinting Large Language Models

Supervised by Prof. Neil Gong

Durham, US

Feb. 2025–Sep. 2025

- Proposed a decision-boundary-based method for constructing model fingerprints
- Designed gray-box and black-box procedures for fingerprint verification
- Demonstrated reliable detection of post-trained and quantized variants of popular models

Alignment of Text-to-Image Models

Supervised by Prof. Neil Gong

Durham, US

Jun. 2024 - Sep. 2024

- Fine-tuned text-to-image models to prevent the generation of harmful content
- Designed a heuristic method to modify the text encoder to generate safe embeddings for unsafe prompts
- Maintained high fidelity of generated images for safe prompts
- Developed defenses to protect open-source models against latest jailbreaking attacks

Removing Image Watermarks from Diffusion Models

Supervised by Prof. Neil Gong

Durham, US

Mar. 2024 - May 2024

- Removed watermarks from open-source watermarked diffusion models while preserving image quality
- Fine-tuned diffusion model decoders with minimal data to achieve high watermark evasion rates
- Removed watermarks like Stable Signature in both encoder-aware and encoder-agnostic scenarios

A Transfer Attack to Image Watermarks

Supervised by Prof. Neil Gong

Durham, US

Oct. 2023 - Feb. 2024

- Developed a transfer attack on watermark-based AI-generated image detectors using multiple surrogate models
- Theoretically analyzed the attack's effectiveness, providing upper and lower bounds for the evasion rate
- Empirically evaluated and compared our method with existing approaches in various scenarios

Model Recovery from Data Poisoning Attacks

Supervised by Prof. Neil Gong

Durham, US

Oct. 2022 - Apr. 2023

- Proposed an efficient method for periodic model recovery from data poisoning attacks
- Designed a diagonalization-based method to efficiently estimate gradients and train recovered models
- Demonstrated certified recovery through theoretical and empirical evaluation for classifiers and recommender systems

PREPRINTS (*EQUAL CONTRIBUTIONS)

MalTool: Malicious Tool Attacks on LLM Agents

- **Yuepeng Hu**, Yuqi Jia, Mengyuan li, Dawn Song, Neil Gong

Log-To-Leak: Prompt Injection Attacks on Tool-Using LLM Agents via Model Context Protocol

- **Yuepeng Hu***, Chongyu Fan*, Sirat Samyoun, Jian Du

SafeText: Safe Text-to-image Models via Aligning the Text Encoder

- **Yuepeng Hu**, Zhengyuan Jiang, Neil Gong

Stable Signature is Unstable: Removing Image Watermarks from Diffusion Models

- **Yuepeng Hu**, Zhengyuan Jiang, Moyang Guo, Neil Gong

PUBLICATIONS (*EQUAL CONTRIBUTIONS)

Fingerprinting LLMs via Prompt Injection

- **Yuepeng Hu**, Zhengyuan Jiang, Mengyuan Li, Osama Ahmed, Zhicong Huang, Cheng Hong, Neil Gong
- Annual Meeting of the Association for Computational Linguistics (ACL Oral), 2026

Leave My Images Alone: Preventing Multi-Modal Large Language Models from Analyzing Images via Visual Prompt Injection

- Zedian Shao*, Hongbin Liu*, **Yuepeng Hu**, Neil Gong
- Annual Meeting of the Association for Computational Linguistics (ACL), 2026

Watermark-Based Detection and Attribution of AI-Generated Content

- Zhengyuan Jiang, Moyang Guo, **Yuepeng Hu**, Neil Gong
- International Conference on Learning Representations (ICLR), 2026

Jailbreaking Safeguarded Text-to-Image Models via Large Language Models

- Zhengyuan Jiang, **Yuepeng Hu**, Yuchen Yang, Yinzhi Cao, Neil Gong
- Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026

A Transfer Attack to Image Watermarks

- **Yuepeng Hu**, Zhengyuan Jiang, Moyang Guo, Neil Gong
- International Conference on Learning Representations (ICLR), 2025

Periodic Recovery from Poisoning Attacks in Machine Learning

- **Yuepeng Hu**, Minghong Fang, Yuqi Jia, Hongbin Liu, Neil Gong
- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025

Webinject: Prompt injection attack to web agents

- Xilong Wang, John Bloch, Zedian Shao, **Yuepeng Hu**, Shuyan Zhou, Neil Gong
- Empirical Methods in Natural Language Processing (EMNLP Main), 2025

Certiably Robust Image Watermark

- Zhengyuan Jiang, Moyang Guo, **Yuepeng Hu**, Neil Gong
- European Conference on Computer Vision (ECCV), 2024

PORE: Provably Robust Recommender Systems Against Data Poisoning Attacks

- Jinyuan Jia, Yupei Liu, **Yuepeng Hu**, Neil Gong
- USENIX Security Symposium, 2023

Screening and Diagnosis of Cardiovascular Disease Using AI-Enabled Cardiac MRI

- Yan-Ran Wang, Kai Yang, Yi Wen*, Pengcheng Wang*, **Yuepeng Hu***, et al.
- Nature Medicine

PROGRAMMING SKILLS

Programming Languages: C, Python, R, Latex